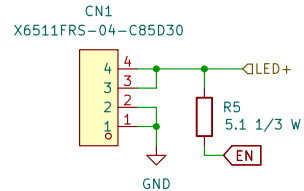
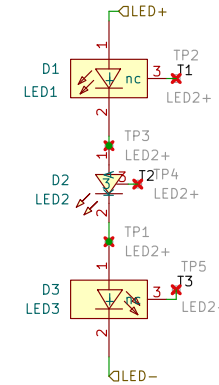


Input voltage: 12 V (3 Led) or 15V (4 Led)
Minimum and maximum voltage depends on the LED configuration



- There are two possible ways to assemble this:
- if external EN or dimming is desired:
 - a) place connector J3
 - b) don't place R5
 - c) use part number BCR421U E6327
 - d) dimming by microcontroller (3.3V)
 - no external EN or dimming:
 - a) don't place J3
 - b) place R5
 - c) use part number BCR420U E6327



Moritz LEDs (all 3535 form factor):

- A) Würth Elektronik Part Numbers:
- Deep Blue (450 nm): 150353DS74500
 - Blue (460 nm): 150353BS74500
 - White (158353060 Cool White)
 - White (158353050 Daylight)
 - White (158353040 Moonlight)
 - Green (150353GS74500 525 nm)
 - Red (150353RS74500 625 nm)
 - UV 365: 15335337AA350
 - UV 395: 15335338AA350
 - UV 395: 15335339AA350
 - UV 405: 15335340AA350
 - Far Red 730: 150353FS74500

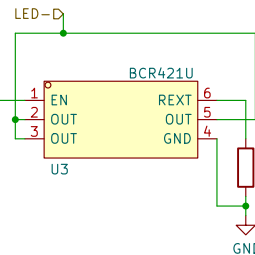
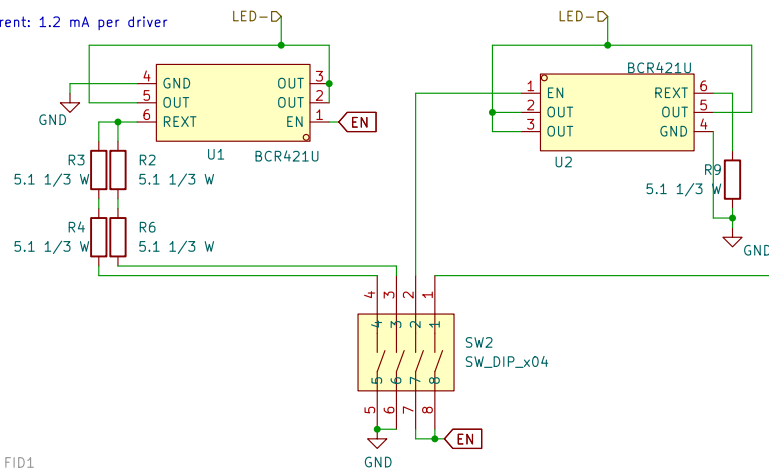
LCSC versions of LEDs

- 365 C2833504 (silverlight brand 700ma)
- 395 C2833576 (silverlight brand 700ma)
- 405 C2833578 (silverlight brand 700ma)
- 450 - C19185871 (Blue 350ma)
- 520 C19185875 (Green 700ma)
- White C19185868 (5500-7000k 700ma)

Testpoints for LEDs are to check for correct soldering and possible shorts below the LEDs between anode, cathode and thermal pad. Also, Vf for individual LEDs can be measured.



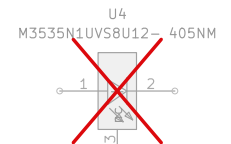
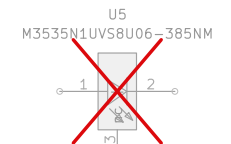
EN current: 1.2 mA per driver



CAUTION - UV

- UV LEDs emit high intensity UV light
- Do not look directly into the UV light during operation. This can be harmful to your eyes and skin.
- Wear protective eyewear to avoid exposure to UV light
- Do not view directly with optical instruments
- Keep out of reach of children

AVOID DIRECT EYE AND SKIN EXPOSURE TO UV LIGHT !



Lab²

LabLab GmbH - modified by DINALAB

Sheet: /
File: Mothbeam.kicad_sch

Title: MothBeam12V

Size: A4 Date: 2026-06-17

KiCad E.D.A. 9.0.9

Rev: 5.0.3

Id: 1/1